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Independent researcher. I build measurement apparatus for language models and spend most of my time on one question: is the signal real, or is the instrument producing it?

That question has cost me two results this year. Both were killed by controls I designed and ran myself, and both are on this page.

CODE — [GITHUB.COM/VEE-INDUSTRIES](https://github.com/Vee-industries)

[Iota-Code](#)

The apparatus, the preregistered controls, and the corpus every number on this page comes from. The measurement scripts run from a clone with no GPU. Apache-2.0.

[detector_memory_glove](#)

The stateful detector layer, its synthetic testbed and the bootstrap intervals below. Apache-2.0.

CURRENT WORK

IOTA — an apparatus for measuring how much of what a language model says next is explained by the conversation so far, rather than by the current prompt. Sole investigator, January 2026 – present. github.com/Vee-industries/Iota-Code

History carries a large, architecture-dependent share of next-output predictability

R is the share of the total predictable information about the emitted token supplied by the conversation history rather than the current prompt — a conditional V-usable information ratio from held-out log-loss under conversation-grouped cross-validation. The collection protocol holds the user prompt byte-identical at a given (run, turn) across every trial and temperature, so the history term can be varied on its own. Checked at 234 of 234 cells, no violations.

LLaMA-3-8B **34.0–55.4%** across five temperatures, lowest bootstrap bound **+27.7%**
Gemma-2-9B **3.8–7.1%** · Mistral-7B-v0.3 **+7.9% to –20.7%**
Excess over permutation null at T0.6 – llama **+0.259**, gemma **+0.071**, Mistral **+0.037** nats
Conversation-cluster bootstrap, B = 300

Zero is not the null for this estimator, and testing against zero inverts conclusions

The predictive family cannot decline to use the history key, so a sparse key costs held-out loss and the no-information expectation is *negative*. Measured against that null instead of against zero, all three architectures are history-dependent and differ in magnitude rather than in kind — including the cells where the point estimate is negative. This is the finding I would most want someone else to take away.

Permutation null **–0.046 to –0.074 nats**, not 0 · history permuted across conversations within each (run, turn) stratum, with targets, prompt key, grouping and both marginal terms held bit-identical

A system prompt telling the model to disengage cuts its history-dependence fourfold

Three runs share one user-prompt sequence and differ only in the system prompt — neutral, cooperative, resistant. The resistant prompt ("deflect abstract questions, do not elaborate") moves R by more than the bootstrap interval, on byte-identical questions.

LLaMA T0.2 – neutral **46.0%** [38.6, 57.5] · cooperative **47.2%** [33.0, 60.0] · resistant **11.7%** [0.6, 20.9] → non-overlapping, 34-point gap

The result survives open generation

Every number above came from an enforced three-way token, which was the apparatus's largest stated limitation. Re-collected the full LLaMA corpus with the enforcer off and the model writing freely; it leads with a status word and continues, so one canonicaliser parses both. The dropped rows were not missing at random — 37.6% of one run against near-zero elsewhere — so that run is excluded and the rest parses at 99.3–100%.

Free generation **31.3 / 40.3 / 40.2 / 35.9 / 48.5%** against enforced **36.4 / 46.1 / 55.6 / 46.6 / 53.5%**
All five intervals overlap · lowest free-generation bound **+23.7%** · free generation reads lower at all five, no single interval separates

Standard attribution credit can go to a control block with negative held-out value

Completing the 2×2 of {model} $\times$ {conversation} showed a donor control block reducing cross-validated R^2 whenever it was actually held out, while Shapley assigned it a sixth of the total — credit accrued in the subsets that lacked the real input block, where it stood in for one.

Unique held-out contribution **negative in 14 of 14 cells** (−0.0002 to −0.0160) · Shapley share **12.9–20.8%** of the same total

WITHDRAWN

Both of these were mine, both were load-bearing, and both were killed by a control that was already inside the experiment. I would rather a reviewer read them here than find them in an old PDF.

WITHDRAWN A three-channel decomposition of the model's next internal state

The original project measured $E + C + R$ over hidden states, with R as the share carried forward from the model's own prior state. It is identically zero by construction: a transformer re-prefills the entire transcript every turn and carries nothing between them. My own no-history control settled it — with history dropped from the input, two cohorts that otherwise differ produce byte-identical hidden states. I retracted the finding, dropped the constraint channel, and redefined the measurand onto the output, where it survives.

With history – across-cohort cosine distance **0.0020141**
History dropped – **0.0**, byte-identical

WITHDRAWN An activation-patching result showing transplanted state beats random noise

Donor hidden states written into the residual stream during generation changed the emitted token far more often than noise did — z from 2.4 to 24.6 at five of six temperatures. The comparison was wrong. The real arm was entirely multi-layer; the noise arm was four-fifths single-layer, and one of those single-layer conditions is inert at 0 of 1300. That one condition generates the entire effect. Run 0017 already contained a properly matched noise arm, and against it the effect vanishes. My own preregistration had required matched layers. The analysis did not honour it.

At matched layers – $z = +0.76, -0.04, -0.52, -0.12, -1.13$ across T0.2–T1.0
Noise numerically higher at four of five · both arms at ceiling at T0.0

PUBLICATIONS

Formal Properties of the Structural Remainder R: An Information-Theoretic Measurement Target

Zenodo · April 2026 · CC BY 4.0 · [10.5281/zenodo.19656944](https://zenodo.org/record/19656944)

Boundedness on $[0, 1]$, invariance under bijective reparameterisation, non-triviality via a closed-form one-bit cache model, identifiability conditions by regime, a closed form in stable linear-Gaussian systems, and monotonicity under two specific erasure operations, with the general class left open. Scoped throughout to *systems with persistent internal state* — state-space models, recurrent networks — and it makes no claim about transformers. The mathematics is unaffected by the withdrawal above; what changed is that full-attention transformers turned out to be the wrong architecture to point it at.

Decomposing Gait Robustness in Demonstration-Free Humanoid Locomotion:
A Predictive Diagnostic for ROM-Guided Reinforcement Learning

Zenodo preprint · April 2026 · CC BY 4.0 · [10.5281/zenodo.19474028](https://zenodo.org/record/19474028)

Four preregistered diagnostic experiments with falsification criteria, applying the same decomposition to locomotion policies — discriminator ablation across training, sensory perturbation at convergence, switching delay under commanded speed change, and an MLP versus LSTM comparison. Proposal only; no experimental results. Written for, and shared with, Dr. Meng Cheng Lau at Laurentian University, whose ROM-guided RL framework it builds on. A recurrent policy carries a hidden state by construction, which makes it one of the architectures where the measure above is well posed.

OTHER WORK

Detector Memory Glove — a stateful severity-scoring layer for intrusion detection

Apart Research / CeSIA AI Incident Response Sprint · September 2026 · github.com/Vee-industries/detector_memory_glove

A per-entity exponential moving average scored alongside the current event, on a synthetic testbed calibrated to the published timing and phase structure of the July 2026 Hugging Face agent intrusion. Includes a diagnostic that locates predictive information in the event stream relative to an external risk label, and an honest account of when the memory advantage is real and when it is an artifact of warm-up.

120-hour review window – detection **94.0%** [93.6, 94.5] with memory against **13.3%** [12.6, 13.9] without · difference **80.8** points [80.0, 81.6]
9,994 attack campaigns · 40,000 benign entities · bootstrap B = 2000 · false-alarm rate 3.7% and 4.9%, both inside the 5% cap

METHODS

ESTIMATION	Conditional V-usable information from held-out log-loss; grouped cross-validation; conversation-cluster bootstrap; permutation nulls; Theil's U as the normalised share. Denominator floors, because four of this project's retractions died by normalising against a quantity that shrinks.
CONTROLS	Preregistration with the decision rule fixed before the run, including arms that came back null and are reported as null. Length-matched scramble controls, a degradation ladder, and a bitwise reproducibility floor across six independent model loads.
CALIBRATION	Built a six-system synthetic calibration suite with analytic ground truth — closed-form truth derivation, an exogeneity-violation sweep, and a deliberate negative control on purely synergistic structure that the estimator failed and that I documented as a failure.
INTERVENTIONS	Activation patching by forward hook into the residual stream, with the emitted token as the outcome; layer-isolation sweeps; ablated-versus-instruct contrasts.
INFRASTRUCTURE	PyTorch, Transformers, NumPy, scikit-learn. 4-bit to FP16 quantisation via bitsandbytes; Hugging Face ecosystem. The full three-family corpus — 6 runs × 13 turns × 40 trials × 5 temperatures — was collected on one 10 GB consumer GPU.
PRACTICE	I specify the architecture, protocol and analysis, and direct the implementation through AI tooling rather than typing it. Every number on this page reproduces from a script in a public repository, and I check them against the files rather than against my notes.

EDUCATION AND EMPLOYMENT

2021	BSc, Interdisciplinary Science , Laurentian University. Quantitative methods, systems analysis, probability and statistics, scientific computing.
2022 –	Passport Officer , Service Canada. Currently on medical leave. Secret clearance (Level II); CSIS security assessment, ten-year background review.
2017 – 2022	Canada Post.
LANGUAGES	English (native), French (working proficiency).

The apparatus, the controls and the measurement scripts are public at github.com/Vee-industries/Iota-Code under Apache-2.0. Every figure here comes from a script in it. Last revised September 2026.